Exp. No: 5

Electric Potential Difference between Two Points in Free Space

Aim:

To write program for Electric potential difference between two points in free space medium using MATLAB.

Objectives:

1. To determine the electric potential difference between two points separated by a distance R in free space.

2. To plot the variation of potential difference with separation distance.

3. To relate the potential gradient to electric field intensity for validation.

PROGRAM

```
clc;
clear;
q = input('Enter the q value: ');
epsilon = 8.854e-12;
k = 1/(4*pi*epsilon);

ax = input('Enter the value of ax = ');
ay = input('Enter the value of ay = ');
r1 = sqrt(ax^2 + ay^2);
v1 = k*q/r1;

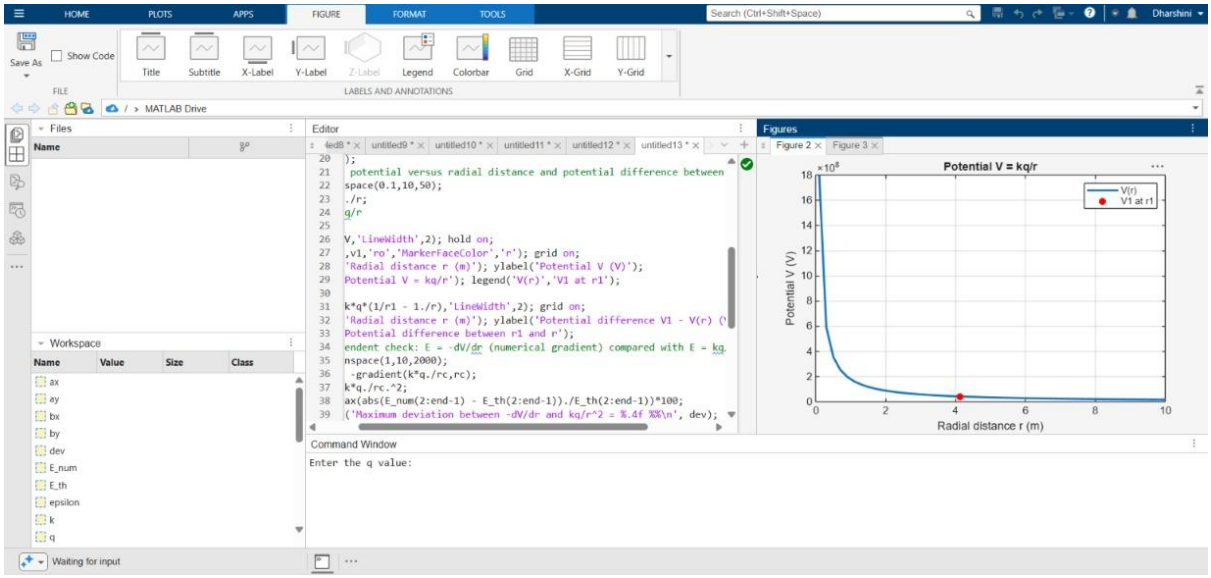
disp('The value of v1');
disp(v1);

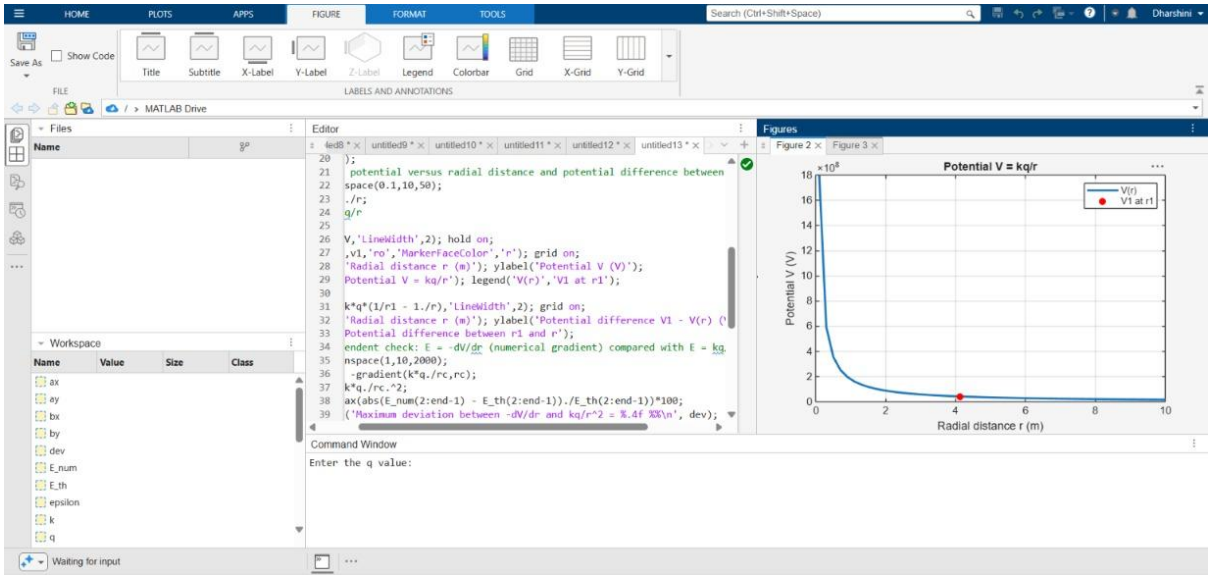
bx = input('Enter the value of bx = ');
by = input('Enter the value of by = ');
r2 = sqrt(bx^2 + by^2);
v2 = k*q/r2;

disp('The value of v2');
disp(v2);

v = v1-v2;
disp('The electric potential difference is');
disp(v);
```

OUTPUT





5(b) Potential Difference for a Positive Charge with Fixed \(r_1\) and Varying \(r_2\)

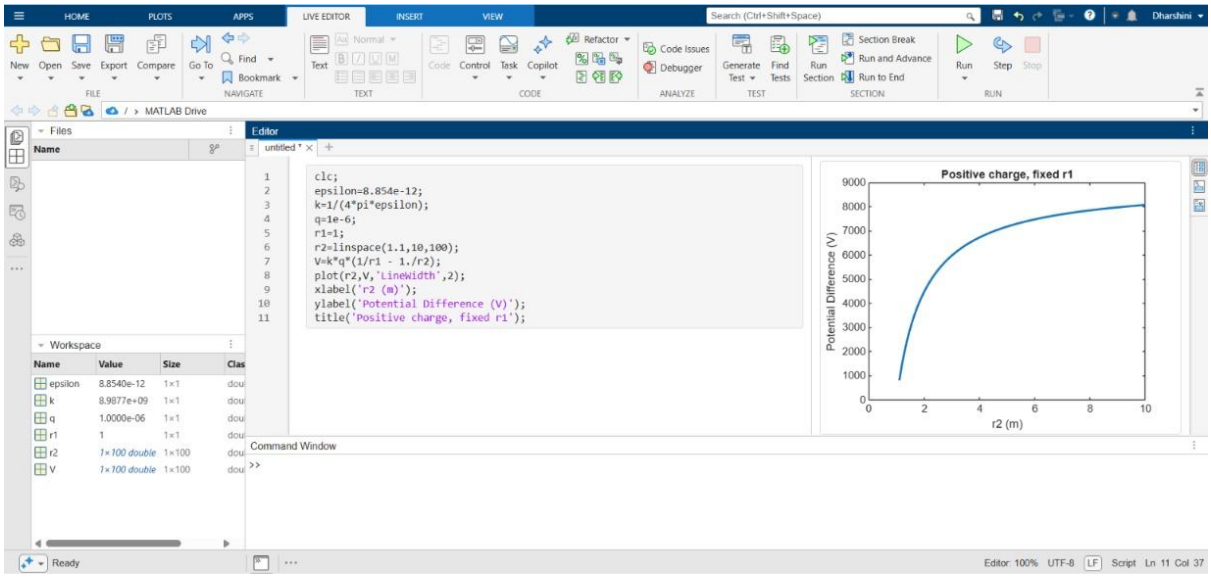
```
clc;
clear;
epsilon = 8.854e-12;
k = 1/(4*pi*epsilon);
q = 1e-6;
r1 = 1;
r2 = linspace(1.1,10,100);

V = k*q*(1/r1 - 1./r2);

plot(r2,V,'LineWidth',2);
grid on;
```

```
xlabel('r2 (m)');
ylabel('Potential Difference (V)');
title('Positive charge, fixed r1');
```

OUTPUT



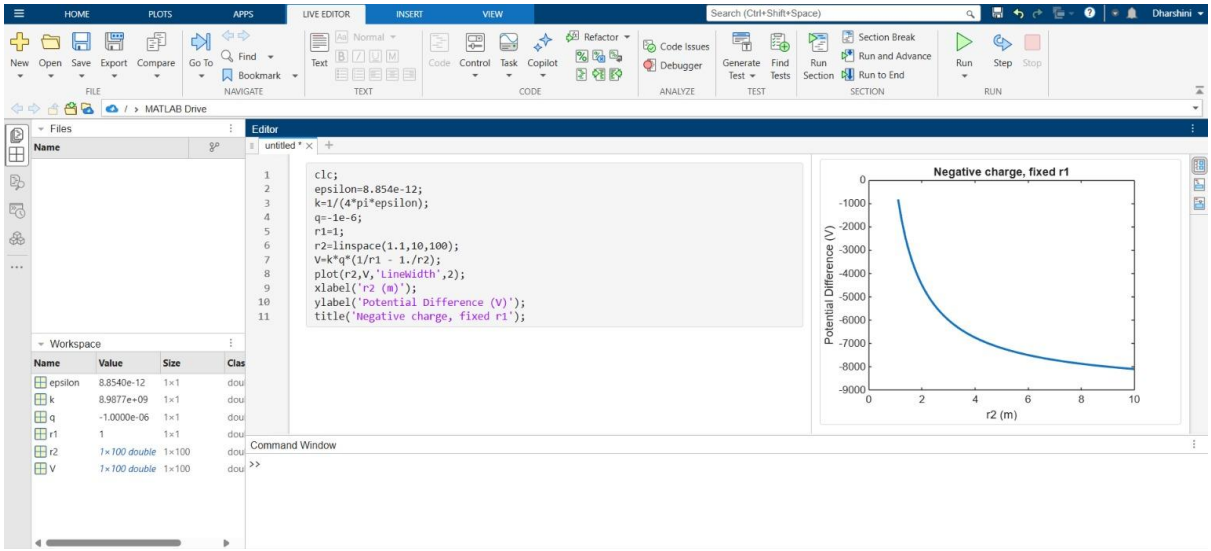
5(c) Potential Difference for a Negative Charge

```
clc;
clear;
epsilon = 8.854e-12;
k = 1/(4*pi*epsilon);
q = -1e-6;
r1 = 1;
r2 = linspace(1.1,10,100);

V = k*q*(1/r1 - 1./r2);

plot(r2,V,'LineWidth',2);
grid on;
xlabel('r2 (m)');
ylabel('Potential Difference (V)');
title('Negative charge, fixed r1');
```

OUTPUT



5(d) Verification of Electric Field Using Potential Gradient

```
clc;
clear;
epsilon = 8.854e-12;
k = 1/(4*pi*epsilon);
q = 1e-6;

r = linspace(1,10,2000);
V = k*q./r;

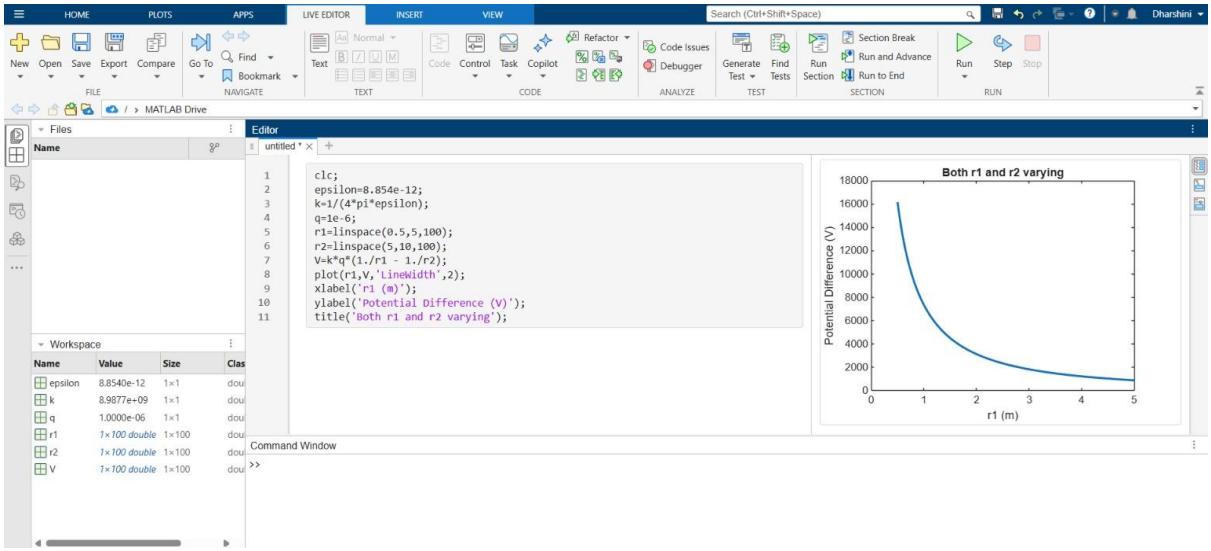
E_num = -gradient(V,r);
E_th = k*q./r.^2;

plot(r,E_num,'b','LineWidth',2);
hold on;
plot(r,E_th,'r--','LineWidth',2);
grid on;
xlabel('Distance (m)');
ylabel('Electric Field (V/m)');
title('Electric Field Verification');
legend('Numerical E','Theoretical E');

dev = max(abs(E_num(2:end-1)-E_th(2:end-1)) ...
    ./abs(E_th(2:end-1)))*100;

fprintf('Maximum deviation = %.4f %%\n',dev);
```

OUPUT


HOME
PLOTS
APPS
LIVE EDITOR
INSERT
VIEW
Search (Ctrl+Shift+Space)
Dharshini
New
Open
Save
Export
Compare
Go To
Find
Bookmark
Text
Normal
Code
Control
Task
Copilot
Refactor
Code Issues
Debugger
Generate Test
Find Tests
Run Section
Section Break
Run and Advance
Run to End
Run
Step
Stop
FILE
NAVIGATE
TEXT
CODE
ANALYZE
TEST
SECTION
RUN
MATLAB Drive
Files
Name
Workspace
Name
Value
Size
epsilon 8.8540e-12 1×1
k 8.9877e+09 1×1
q 1.0000e-06 1×1
r1 1×100 double 1×100
r2 1×100 double 1×100
V 1×100 double 1×100
Editor
untitled *
1 clc;
2 epsilon=8.854e-12;
3 k=1/(4*pi*epsilon);
4 q=1e-6;
5 r1=linspace(0.5,5,100);
6 r2=linspace(5,10,100);
7 V=k*q*(1./r1 - 1./r2);
8 plot(r1,V,'LineWidth',2);
9 xlabel('r1 (m)');
10 ylabel('Potential Difference (V)');
11 title('Both r1 and r2 varying');
Both r1 and r2 varying
Potential Difference (V)
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0
1
2
3
4
5
r1 (m)
Command Window
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